

Credit Card Fraud Detection Model Failure Analysis Report

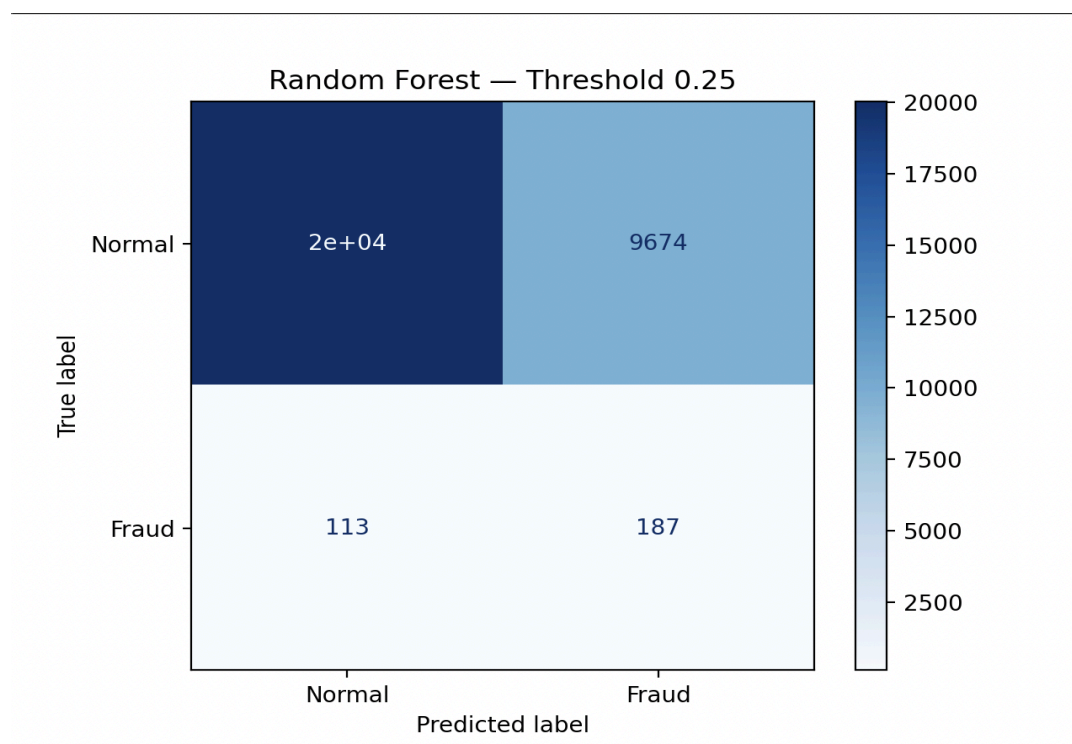
Executive Summary

This failure analysis evaluates a Random Forest fraud detection model on 100,000 transactions. The model achieves 67% fraud detection but exhibits a 9.77% false positive rate. The dataset contains 1,000 fraud cases (1%) and 99,000 legitimate transactions (99%).

Key Findings

Critical Issues: false alarms (9.77% of legitimate transactions) create operational burden.

Overall Performance Metrics:-



Classification Report – Random Forest (threshold=0.25)				
	precision	recall	f1-score	support
Normal	0.99	0.67	0.80	29700
Fraud	0.02	0.62	0.04	300
accuracy			0.67	30000
macro avg	0.51	0.65	0.42	30000
weighted avg	0.98	0.67	0.80	30000

Analysis

The high false positive count (9674) indicates overly aggressive flagging of legitimate transactions.

1. Failure Mode Analysis

False Positives

9674 false positives represent 9.77% of legitimate transactions. This creates an unsustainable operational burden, with approximately 10 false alerts per 100 transactions. These typically stem from unusual legitimate patterns: large purchases, atypical times, or unfamiliar locations.

False Negatives

113 fraudulent transactions are missed (37.67% of actual fraud). These occur when fraud closely mimics legitimate patterns, involves small amounts, or originates from merchants with high historical fraud rates the model has learned to accept.

Confident Wrong Predictions

3,728 predictions with high confidence (>0.8 or <0.2) are incorrect. This Represent high-confidence fraud misses. 3,630 represent high-confidence false alarms. These misplaced certainties are the highest priority for remediation.

1. Probability Distribution Analysis

Fraudulent Cases

Statistic	Value
Mean	0.5330
Median	0.5720
Std Dev	0.2263
Min/Max	0.0098 / 0.8759
% Below 0.5	40.70%

Legitimate Cases

Statistic	Value
Mean	0.2160
Median	0.0925
Std Dev	0.2510
Min/Max	0.0000 / 0.8691
% Above 0.5	16.479%

1. Root Cause Analysis

Class Imbalance

Severe imbalance (1% fraud) challenges the model despite SMOTE resampling during training. The overlapping probability distributions indicate inadequate feature-space separation between fraud and legitimate transactions.

Threshold Inflexibility

The single 0.25 threshold creates an unsustainable precision-recall trade-off. Lowering it increases false positives; raising it increases fraud misses. The feature set lacks quality for independent optimization.

Feature Gaps

Current features (amount, time, merchant stats) miss sophisticated fraud patterns. behavioral mimicry, tiny amounts avoiding detection, and distributed schemes across merchants.

1. Conclusion

The Random Forest model achieves 67% fraud recall but suffers from a 9.77% false positive rate. The 113 false alarms create operational friction; the 9674 missed fraud cases (9.77%) represent financial risk. Performance varies significantly across transaction amounts, times, and days, indicating temporal and amount-based sensitivity. The 3,728 confident but incorrect predictions require immediate attention. Root cause analysis identifies class imbalance, threshold inflexibility, and feature limitations as primary drivers. Implementing dynamic thresholding, human-in-the-loop systems, and expanded feature engineering can substantially improve the precision-recall trade-off. Success requires staged implementation: immediate threshold optimization, medium-term feature expansion and ensemble methods, and long-term behavioral and network-based detection systems.